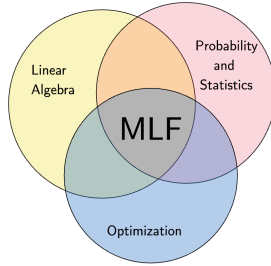


Week-3, Session-2



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Disclaimer: These notes were generated as a part of this [live session](#). It can't be used as a standalone resource or a set of notes for this week.

Whatever is marked in a box indicating "**Error Alert**" indicates an error that I made during a session. **red** indicates the erroneous statement. The correction to that statement is provided in **green** immediately below it.

1. System of Linear Equations

There are m linear equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

Now if you collect the coefficients in a matrix and the RHS and the unknowns in vectors:

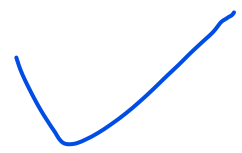
$$Ax = b$$

- A is of shape $m \times n$
- $x \in \mathbb{R}^n$
- $b \in \mathbb{R}^m$

One can look at this system in two ways. After professor Gilbert Strang, we have the row picture and the column picture.

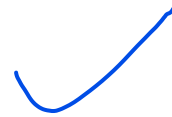
1.1. Row picture

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$



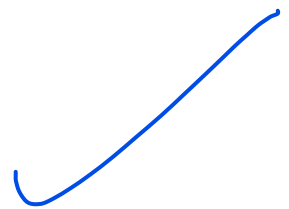
You can treat each row of A as a row vector. The i^{th} row is $a_i^T \in \mathbb{R}^n$.

$$\begin{array}{l} a_1^T x = b_1 \\ \vdots \\ a_m^T x = b_m \end{array}$$



1.2. Column picture

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$



When A is expressed as a collection of n column-vectors, then Ax is the linear combination of the columns of A . Therefore, the LHS is a linear combination of columns.

$$x_1 a_1 + \cdots + x_n a_n = b$$

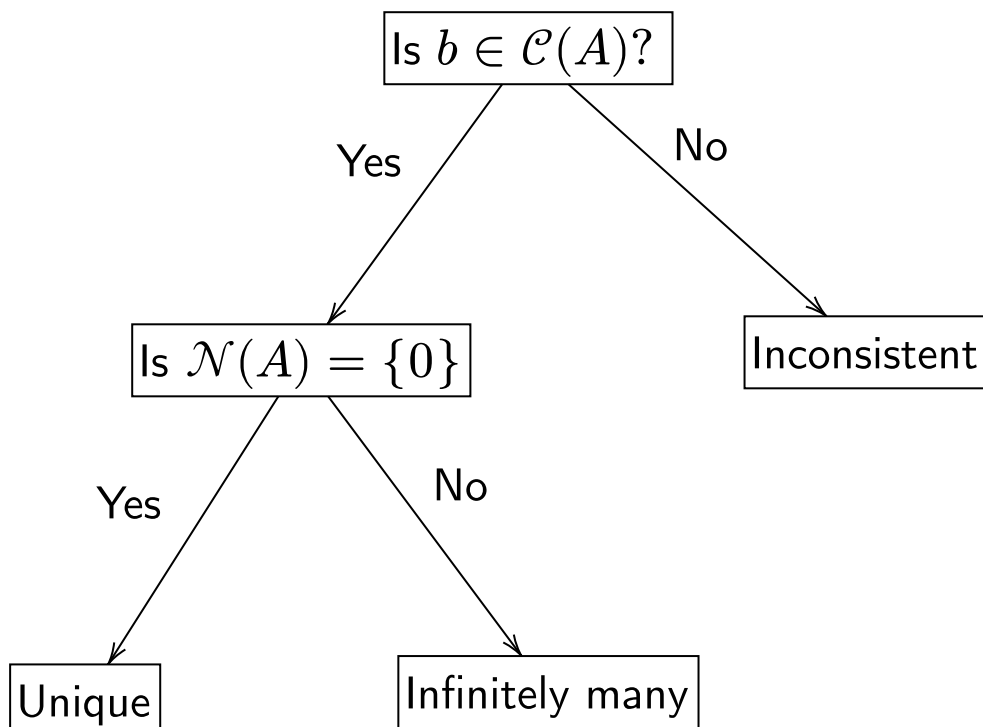
If $Ax = b$ should have a solution, if it is to be consistent, that is, then b should be in the column space of A . Given its importance, we make it more prominent:

Necessary and sufficient condition for consistency: A system of linear equations, $Ax = b$, is consistent if and only if b is in $\mathcal{C}(A)$, where $\mathcal{C}(A)$ is the column space of A .

Recall that a system of linear equations is consistent if it has at least one solution. The following is an example of a system m that is **not consistent**.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

We can come up with the following tree. Refer to the remainder of the live session to understand how this tree is constructed.



2. $Ax = 0$, homogeneous system

Quickly, the algorithm:

- Find the RREF of A
- Identify pivots and pivot columns
- Identify free variables and pivot variables
- Follow the usual procedure; refer to this [document](#) for the full details

The set of all solutions to $Ax = 0$ is called the nullspace of A and denoted by $\mathcal{N}(A)$.

- $\mathcal{N}(A)$ is subspace of $\mathbb{R}^n$
- If $Ax = 0$ has a unique solution, then $\mathcal{N}(A) = \{0\}$
- The following are equivalent to the statement that $\mathcal{N}(A) = \{0\}$
 - The columns are linearly independent.
 - There are no free (independent) variables while solving $Ax = 0$
 - There are n pivot variables when solving $Ax = 0$
 - The matrix has full column rank

- **Warning:** If in addition the matrix is A square, then this will happen when $|A| \neq 0$ or when A is invertible.

Error Alert:

The following erroneous statements were made in the live session.

If the columns span $\mathbb{R}^n$ then $\mathcal{N}(A) = \{0\}$.

If the columns form a basis for $\mathbb{R}^n$ then $\mathcal{N}(A) = \{0\}$

- The first error is that the columns are in $\mathbb{R}^m$ and not $\mathbb{R}^n$.
- Secondly, it is sufficient for the columns to be linearly independent. It is not necessary that they span $\mathbb{R}^m$ for the nullspace to be $\{0\}$ or that they are a basis for $\mathbb{R}^m$.
- These statements are true only if $m = n$.

3. $Ax = b$

The goal is to find the set of all solutions to $Ax = b$ assuming that it is consistent.

Check later:

$$(1, 2, 0, 1)$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

Step-1: Augmented matrix

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 2 & 1 & 1 & 1 & 5 \\ 1 & 2 & 1 & 1 & 6 \end{bmatrix}$$

Step-2: Row reduction

Also called forward elimination.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 2 & 1 & 1 & 1 & 5 \\ 1 & 2 & 1 & 1 & 6 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 1 & 2 & 1 & 1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 1 & 2 & 1 & 1 & 6 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 0 & 2 & -1 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 0 & 2 & -1 & 1 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 0 & 0 & 5 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 0 & 0 & 5 & -1 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow \frac{R_3}{5}} \begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 1 & 3 \\ 0 & 0 & 1 & -1/5 & -1/5 \end{bmatrix}$$

Step-3: Identify the pivots and the pivot columns

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & b \\ \mathbf{1} & 0 & 2 & 0 & 1 \\ 0 & \mathbf{1} & -3 & 1 & 3 \\ 0 & 0 & \mathbf{1} & -1/5 & -1/5 \end{bmatrix}$$

Step-4: Identify free and pivot variables

- Free variables: x_4
- Pivot variables: x_1, x_2, x_3

Step-5: Identify the nature of solutions

Since there is a free variable, the system has infinitely many solutions.

Step-6: Particular solution

The easiest way to find one solution is to set the free variables to zero and solve for the pivot variables using **backward substitution**.

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & b \\ \mathbf{1} & 0 & 2 & 0 & 1 \\ 0 & \mathbf{1} & -3 & 1 & 3 \\ 0 & 0 & \mathbf{1} & -1/5 & -1/5 \end{bmatrix}$$

$$x_1 = 1 - 2x_3$$

$$= 1 + \frac{2}{5}$$

$$= \frac{7}{5}$$

$$x_2 = 3 + 3x_3$$

$$= 3 - \frac{3}{5}$$

$$= \frac{12}{5}$$

$$x_3 = -1/5$$

$$x_4 = 0$$

Therefore, a solution to $Ax = b$ is:

$$x_p = \left(\frac{7}{5}, \frac{12}{5}, \frac{-1}{5}, 0 \right)$$

Error Alert:

I multiplied by 5 and called that a particular solution.

When solving $Ax = b$, one can't arbitrarily scale a solution. This makes sense only while solving $Ax = 0$, the homogeneous system.

x_p is called a particular solution to the system. Note that $Ax_p = b$.

What is the general solution?

If $x_n \in \mathcal{N}(A)$, then $Ax_n = 0$. What can you say about $x_p + x_n$? Is that a solution to the system $Ax = b$?

$$\begin{aligned} A(x_n + x_p) &= Ax_n + Ax_p \\ &= 0 + b \\ &= b \end{aligned}$$

Therefore, $x_n + x_p$ is a solution to $Ax = b$. One can show that:

Let $Ax = b$ be consistent and let x_p a particular solution. Then the set of all solution to $Ax = b$ is:

$$x_p + \mathcal{N}(A)$$

That is, any solution to $Ax = b$ can be written as a particular solution plus some vector from the nullspace of A .

If $Ax = b$ is consistent, when does $Ax = b$ have a unique solution?

- If the nullspace is $\{0\}$
- Refer to $Ax = 0$ discussion for other ways of representing this.

Remark: If A is invertible, then there is a unique solution and that is given as $x = A^{-1}b$.

4. $Ax \approx b$

$Ax = b$ is not always consistent, especially in a practical setting. In such cases, the best thing to do is to resort to an approximation.

Approximating scalar values, say $\sqrt{2}$

- 1.414
- 1.41
- 1.4

Goodness of approximation:

$$e = \sqrt{2} - 1.414$$

Smaller the error, better the approximation.

We now have an inconsistent system, therefore, we need to look for approximations:

$$Ax \approx b$$

Can I get close to b ? Can I find an x that will get me as close to b as possible?

$$b = (1, 2, 3)$$

$$Ax' = (1.5, 2.5, 3.5)$$

$$Ax'' = (1.1, 2.1, 3.1)$$

For approximating vectors, what process are you going to follow?

We could use simple differences as a measure of the goodness of the approximation:

$$e' = (1.5 - 1) + (2.5 - 2) + (3.5 - 3)$$

But that has a problem as the differences might cancel each other out.

$$(1.5, 2, 2.5)$$

$$(1.5 - 1) + (2 - 2) + (2.5 - 3) = 0$$

Therefore, take the square of the differences:

$$e = (1.5 - 1)^2 + (2.5 - 2)^2 + (3.5 - 3)^2$$

The sum of squared errors (SSE).

5. Linear Regression and Least Squares

Consider the following problem:

$$\begin{bmatrix} Q_1 & Q_2 \\ 50 & 70 \\ 60 & 60 \\ 70 & 80 \\ 100 & 90 \end{bmatrix} \begin{bmatrix} E \\ 80 \\ 60 \\ 60 \\ 80 \end{bmatrix}$$

Based on historical data of performance of previous term students in MLF, you as an admin want to predict how students in the current term will perform in the end-term based on their Q1, Q2 scores.

This is an example of a regression problem: you want to predict a real number.

- Features: Q1 and Q2 marks
- Label (target): end-term marks

If the model that you choose to represent the relationship between the labels and the features is linear, then you have linear regression.

$$\hat{E} = x_0 + x_1 Q_1 + x_2 Q_2$$

$$50x_1 + 70x_2 \approx 80$$

$$60x_1 + 60x_2 \approx 60$$

$$70x_1 + 80x_2 \approx 60$$

$$100x_1 + 90x_2 \approx 80$$

This is an overdetermined system. More equations than unknowns.

$$\begin{bmatrix} 50 & 70 \\ 60 & 60 \\ 70 & 80 \\ 100 & 90 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \approx \begin{bmatrix} 80 \\ 60 \\ 60 \\ 80 \end{bmatrix}$$

$$Ax \approx b$$

- A holds the features of your data (comes from history)
 - The first column is Q1 scores
 - The second column is Q2 scores
- b holds the true labels (comes from history)
- x holds the weights
 - x_1 gives the importance of Q1 in predicting the end-term score (label)
 - x_2 gives the importance of Q2 in predicting the end-term score (label)

Find x such that Ax is as close to b as possible. Your prediction should be as close to the target as possible. You can define the error in prediction as:

$$\begin{aligned} e_1^2 &= (50x_1 + 70x_2 - 80)^2 \\ &\vdots \\ e_4^2 &= (100x_1 + 90x_2 - 80)^2 \end{aligned}$$

You therefore have the SSE which you have to minimize

$$\min_{x_1, x_2} e_1^2 + \dots + e_4^2$$

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

You can express the predicted scores as $a_i^T x$. The SSE, often called the loss function (error function).

$$L(x) = (a_1^T x - b_1)^2 + \dots + (a_m^T x - b_m)^2$$

Your goal is to:

$$\min_x L(x)$$

- $L(x)$ is called the objective function
- Unconstrained optimization (week-8)

6. Normal equations

Solution strategy:

- Differentiate
- Set derivative to zero

$L(x)$ is a function of x , where $x = (x_1, \dots, x_n)$. Therefore, you have to compute the gradient and set it to zero.

$$\nabla L(x) = \begin{bmatrix} \frac{\partial L}{\partial x_1} \\ \vdots \\ \frac{\partial L}{\partial x_n} \end{bmatrix}$$

$$L(x) = (a_1^T x - b_1)^2 + \dots + (a_m^T x - b_m)^2$$

$$\nabla (a_i^T x - b_i)^2 = 2(a_i^T x - b_i) a_i$$

$$\frac{1}{2} \nabla L(x) = (a_1^T x - b_1) a_1 + \dots + (a_m^T x - b_m) a_m$$

$$a_1 (a_1^T x - b_1) + \dots + a_m (a_m^T x - b_m)$$

$$a_1 a_1^T x - a_1 b_1 + \dots + a_m a_m^T x - a_m b_m$$

- a_i is a vector
- x is a vector
- b_i is a scalar

$$\frac{1}{2} \nabla L(x) = (a_1 a_1^T + \dots + a_m a_m^T) x - [b_1 a_1 + \dots + b_m a_m] \quad (1)$$

The product of two matrices can be written as the sum of rank-1 matrices:

$$A^T = \begin{bmatrix} | & & | \\ a_1 & \dots & a_m \\ | & & | \end{bmatrix}, \quad A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

$$A^T A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_m \\ | & & | \end{bmatrix} \begin{bmatrix} - & a_1^T & - \\ \vdots & & \\ - & a_m^T & - \end{bmatrix}$$

$$A^T A = a_1 a_1^T + \cdots + a_m a_m^T$$

$$A^T b = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_m \\ | & & | \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = b_1 a_1 + \cdots + b_m a_m$$

$$\frac{1}{2} \nabla L(x) = (A^T A)x - A^T b \quad (2)$$

Now setting the gradient to zero:

$$\boxed{(A^T A)x = A^T b}$$

These are the called the normal equations. Solving this will give you the solution to what is called the least squares problem. The solution is called the least squares solution. This was pioneered by Gauss in astronomy.

- This is also a system of linear equations:
 - $A^T A$ is of shape $n \times n$
 - $x \in \mathbb{R}^n$
 - $A^T b \in \mathbb{R}^n$
 - * A^T is $n \times m$
 - * $b \in \mathbb{R}^m$
- What is the guarantee that $(A^T A)x = A^T b$ is consistent?
 - One can show that this system is always consistent
- When does $(A^T A)x = A^T b$ have a unique solution?
 - When $\mathcal{N}(A) = \{0\}$
 - $A^T A$ is invertible provided A has full column rank

For example:

$$\begin{bmatrix} 50 & 70 \\ 60 & 60 \\ 70 & 80 \\ 100 & 90 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \approx \begin{bmatrix} 80 \\ 60 \\ 60 \\ 80 \end{bmatrix}$$

Step-1: Find $A^T A$

Step-2: $A^T b$

Step-3: Find the augmented matrix

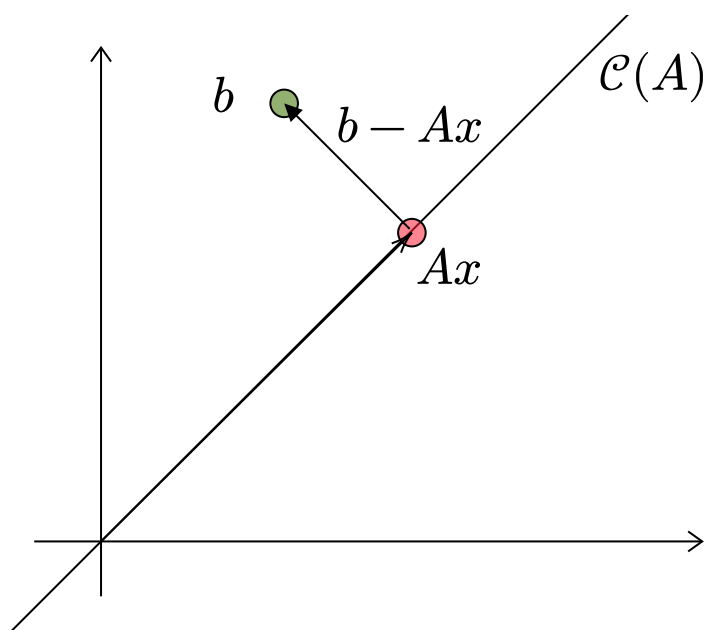
Step-4: Do row reduction and solve for x , find some particular solution

7. Projections

We are trying to find a vector Ax that is as close to b as possible. What is Ax ? It is a linear combination of the columns of A . Therefore, Ax is some vector in the column space of A .

Find a vector in the column space of A that is closest to b

Recall that $\mathcal{C}(A)$ is a subspace of $\mathbb{R}^m$ and $b \in \mathbb{R}^m$



Observe that $b - Ax$ is perpendicular to the column space of A . This means $b - Ax$ is perpendicular to every column of A . The columns of A

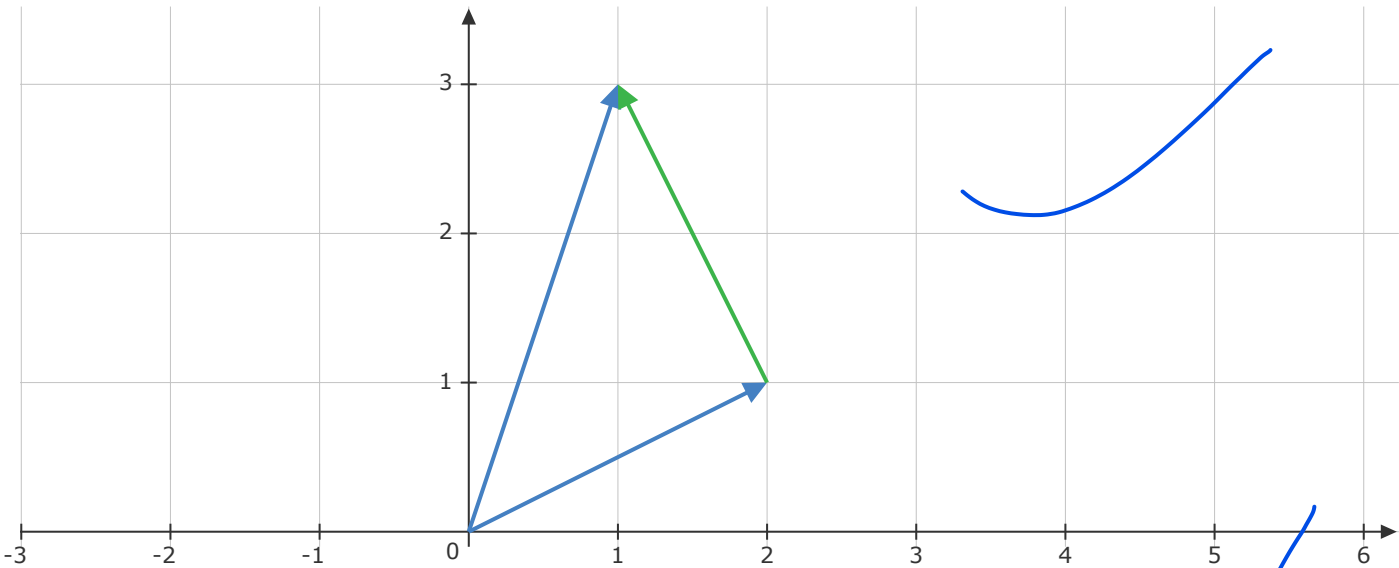
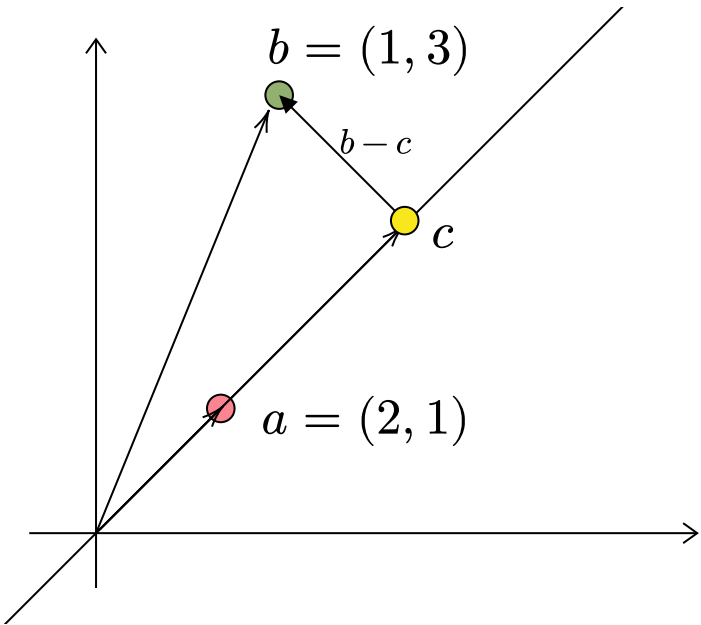
are the rows of A^T

$$A^T(b - Ax) = 0$$

$$A^Tb = A^T Ax$$

$$(A^T A)x = A^T b$$

Geometrically, the least squares solution is trying to find the orthogonal projection of b onto column space of A . The resulting projection Ax .



$$a \perp (b - c)$$

$$a^T(b - c) = 0$$

$$a^T(b - \alpha a) = 0$$

Scalar projection

$$\alpha = \frac{a^T b}{a^T a}$$

The projection of b onto the line given by the vector a :

$$\frac{a^T b}{a^T a} \cdot a = \frac{a^T b}{||a||^2} \cdot a = \left(\frac{a}{||a||} \right)^T b \cdot \left(\frac{a}{||a||} \right)$$

$$b = (1, 3), \quad a = (2, 1)$$

$$\frac{5}{5} \cdot (2, 1)$$

In general, the projection of a vector b onto a vector a is given by:

$$\frac{b^T a}{a^T a} a$$

If the vector a happens to have unit norm, meaning $||a||^2 = a^T a = 1$, then, the projection of b onto a can be expressed as:

$$(b^T a) a$$

8. Projection matrices

Projection of b onto the line a :

$$\frac{b^T a}{a^T a} a = a \frac{a^T b}{a^T a} = \left(\frac{1}{a^T a} \cdot a a^T \right) b$$

$$P = \frac{a a^T}{a^T a}$$

Therefore, the projection of b onto a is given by Pb :

$$b \rightarrow Pb$$

You can verify that $b - Pb \perp a$

$$a^T \left(b - \frac{a a^T b}{a^T a} \right) = a^T b - \frac{a^T a}{a^T a} \cdot a^T b = 0$$

$$P^2 = \frac{1}{(a^T a)^2} a a^T a a^T = \frac{a a^T}{a^T a} = P$$

If $a \in \mathbb{R}^n$, then P is of shape $n \times n$. P projects vectors in $\mathbb{R}^n$ onto the line which passes through the vector a .

In general, you can find a projection matrix for any subspace of $\mathbb{R}^n$.

Projection matrices satisfy the following properties:

- $P^2 = P$
- P is symmetric